



ST BENEDICT'S COLLEGE

SUBJECT	LIFE SCIENCES	DATE	2 OCTOBER 2023
GRADE	11	MARKS	40
EXAMINER	MRS S. STEGMANN	MODERATOR	MRS B. ZAJAC
NAME		DURATION	40 MINUTES
CLASS			

INSTRUCTIONS:

1. This paper consists of 8 pages- including 1 page of additional writing space. Please check that your question paper is complete.
 2. Answer all questions **on the question paper**.
 3. Write in blue or black pen only.
 4. Read all questions carefully and answer in as much detail as possible.
 5. All diagrams should have a clear heading.
 6. It is in your own interest to write legibly and to present your work neatly.
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QUESTION	POSSIBLE MARK	MARK ACHIEVED
1	10	
2	13	
3	17	
TOTAL: 40		

Question 1:

1. Select the name/ or term in Column II which best matches the description in Column I. Write the letter of the matching name/term in the appropriate space provided between **the brackets**. Each letter may only be used once. [10]

Column I

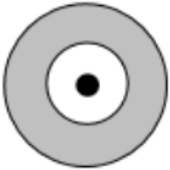
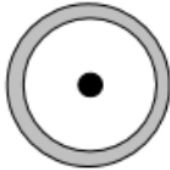
- [] Antibodies
- [] Viruses that infect bacteria
- [] Weakened immune system
- [] Algae
- [] Germs
- [] Agar
- [] Acellular
- [] Mosquito
- [] Antibiotics
- [] Thrush

Column II

- A** Gel extract
- B** HIV
- C** Pathogens
- D** Fungus
- E** B lymphocytes
- F** Vector
- G** Protist
- H** Bacteriophage
- I** Bacteria
- J** Virus
- K** T lymphocytes
- L** decomposers

Question 2

Valentina and Ishika are microbiology students investigating penicillin resistance in modern day bacterial cultures. Penicillin had, in the past, been an effective antibiotic and so they decided to set up an experiment to determine which type of bacterium has developed the greatest resistance to penicillin. They obtained pure cultures of five different types of bacteria (A-E) and placed each type of bacteria into a separate petri dish with nutrient agar. Once the agar was set, a hollow was cut in the agar in the center of each petri dish, using a cork borer. Penicillin was then added to the hollows. After incubation at 25°C for 24 hours, the petri dishes were examined.

KEY					
Area with living bacteria			Area containing penicillin		
			Clear zone – no bacteria		
Start of experiment	Bacteria Type A	Bacteria Type B	Bacteria Type C	Bacteria Type D	Bacteria Type E
					
End of experiment					

2.1. State the independent variable in this experiment. (1)

2.2. Write a suitable hypothesis for the experiment. (2)

2.3. Which bacteria showed the most antibiotic resistance? (1)

2.4. Explain your answer to QUESTION 2.3. (2)

2.5. Explain how resistance to antibiotics develops in a population of bacteria. (3)

2.6. State ONE way in which the development of resistant bacteria can be minimized. (1)

2.7. Valentina and Ishika forgot to set up controls for this experiment.

A. Explain why controls are necessary in this experiment. (2)

B. Which 'ingredient' would be missing from the controls? (1)

[13]

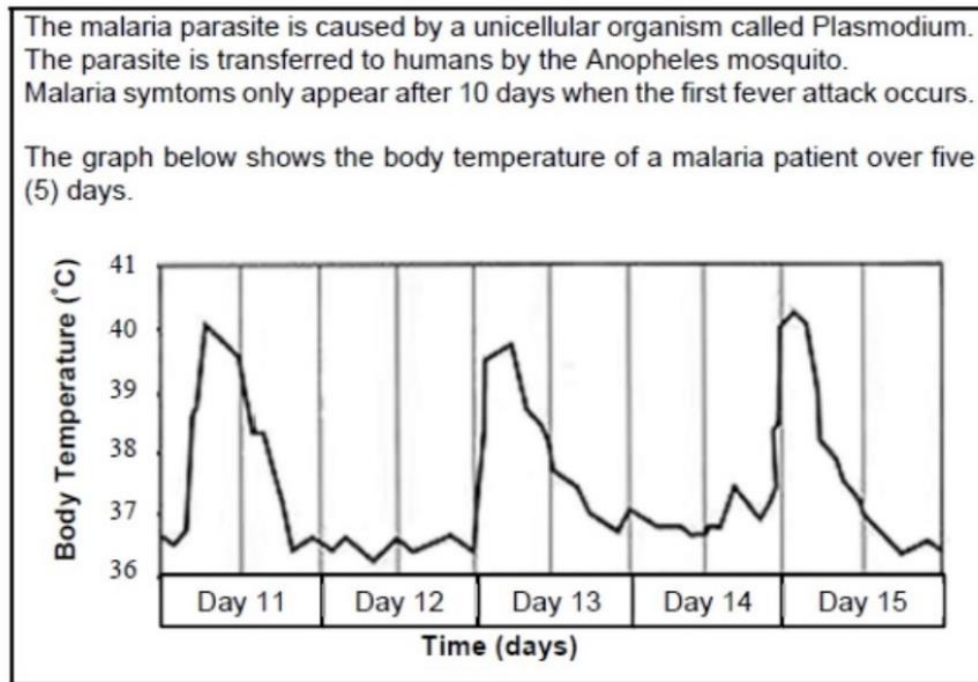
Question 3

- 3.1. Read the extract below and study the graph showing the temperature of a patient suffering with malaria.

The malaria parasite is caused by a unicellular organism called Plasmodium. The parasite is transferred to humans by the Anopheles mosquito.

Malaria symptoms only appear after 10 days when the first fever attack occurs.

The graph below shows the body temperature of a malaria patient over five days.



- 3.1.1. What group of microorganisms does Plasmodium belong to? (1)

- 3.1.2. From the chart determine:

- a. The highest temperature this patient experienced during the time period. (1)

- b. The number of fever attacks the patient experienced during the five days. (1)

- 3.1.3. Why does the graph above begin on Day 11? (1)

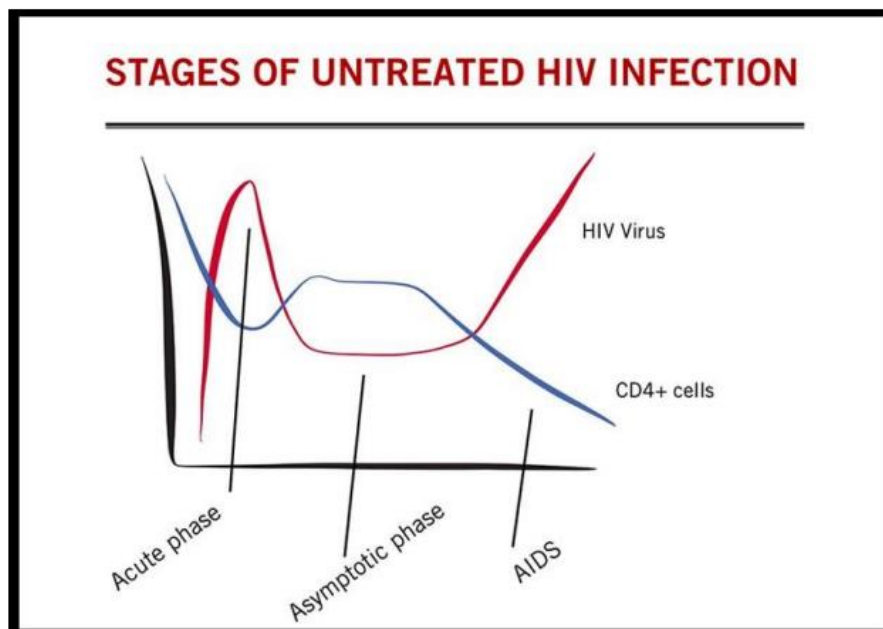
3.1.4. Predict on which day the patient could expect the start of the next fever attack. (1)

3.1.5. *Plasmodium* destroys the red blood cells. What effect could this have on the body? (2)

3.1.6. Suggest TWO ways of preventing malaria. (2)

[9]

3.2. Acute HIV infection is the earliest stage of HIV infection, and it generally develops withing 2 to 4 weeks after infection with HIV. During this time, some people have flu-like symptoms, such as fever, headache, and rash. In the acute stage of infection, HIV multiplies rapidly and spreads throughout the body.



3.2.1. Explain the function of CD4 cells. (1)

3.2.2. When will a person be classified with AIDS? (2)

3.2.3. Explain why the HIV virus is so dangerous and causes so many deaths. (3)

3.2.4. Why is it possible to easily cultivate and grow bacteria, but it is very difficult to cultivate a virus? (2)

[8]

[17]

TOTAL: 40 MARKS

[illegible]